

Sparse Biomarker Discovery of tDCS Treatment Response in Adults with Treatment-Resistant Depression via Low Montage Resting-State EEG Analysis

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ABSTRACT

We evaluated whether a sparse, low-montage resting-state EEG pipeline could identify clinically interpretable biomarkers of response to transcranial direct current stimulation (tDCS) in adults with treatment-resistant depression under strict patient-level validation. The study used 4-channel eyes-closed EEG and a controlled sweep of 150 successful runs across two model families: an advanced hybrid 1D CNN-LSTM and a linear support vector machine (SVM). The sweep varied window size (2–10 s) and the per-fold feature-selection budget (inner-k 1–70), while enabling both leave-one-participant-out group equalization and SMOTE during training.

The strongest configuration was the hybrid model with 6 s windows and inner-k=1. That run reached patient-level ROC-AUC 0.816, PR-AUC 0.579, balanced accuracy 0.786, accuracy 0.810, and MCC 0.571 over 21 held-out participants. The best SVM configuration (8 s, inner-k=30) reached ROC-AUC 0.510 and MCC 0.277. The hybrid winner was also highly sparse, selecting 1 feature per fold and only 7 unique features overall, with a permutation-tested ROC-AUC advantage over chance ($p = 0.008$). Hard-set overlap across folds was modest (mean pairwise Jaccard 0.190; Kuncheva index 0.187). The recurrent features showed perfect mean effect-direction consistency.

These results support a sparse short-timescale EEG framework that recovers a small, interpretable biomarker set while preserving useful patient-level discrimination in a relatively small cohort. The most recurrent features involved temporal spectral entropy and frontal-temporal asymmetry-derived measures, and their class-conditional selection patterns motivate follow-up validation in larger externally evaluated datasets.

1. Introduction

Treatment-resistant depression (TRD) remains a major clinical challenge (Milev et al., 2016; Mutz et al., 2018; Razza et al., 2020). Neuromodulation approaches such as transcranial direct current stimulation (tDCS) continue to attract interest as potentially scalable adjunctive interventions (Boggio et al., 2007; Brunoni et al., 2016; Thair et al., 2017). Broader translational neuromodulation work has reinforced that clinical interest (Cramer et al., 2011). More recent studies have pushed tDCS into remotely supervised and home-based settings, which makes pretreatment stratification a practical clinical problem (Woodham et al., 2025; Rimmer et al., 2024). A useful biomarker pipeline has to identify which patients are more or less likely to respond before stimulation begins.

In that setting, EEG is appealing because it is comparatively inexpensive, portable, and already used across translational neurophysiology workflows (Casson et al., 2010; Miranda et al., 2019; Simmatis et al., 2023). Wearable and reduced-channel EEG systems further increase deployment feasibility, although they also sharpen the trade-off between practicality and signal richness (Siuly et al., 2017; BABILONI et al., 2012; Stoyell et al., 2021).

The literature on EEG biomarkers in major depressive disorder is encouraging and methodologically uneven. Reviews and biomarker overviews have reported potentially informative spectral, asymmetry, vigilance, entropy, and complexity signatures (Olbrich and Arns, 2013; Olbrich et al., 2015; Leuchter et al., 2010). Related syntheses have focused more directly on treatment response and clinical translation (Fu and Costafreda, 2013; Fu et al., 2019; Widge et al., 2019). Predictive-marker overviews and technical guidance reinforce the same standardization pressures (Watts et al., 2022; Cohen et al., 2023; Simmatis et al., 2023). Machine learning studies have extended this theme into antidepressant, placebo, and neuromodulation response prediction. External generalization, feature stability, leakage-safe feature selection, and interpretability remain active concerns, especially when relatively small cohorts are paired with high-dimensional feature spaces (Al-Kaysi et al., 2017; Jaworska et al., 2019; Oakley et al., 2022). Related nonlinear and entropy-oriented studies point to the same caution about dimensionality and signal instability (Shalbaf et al., 2018; Ebrahimzadeh et al., 2021; Zhdanov et al., 2020). Entropy-focused EEG work and recent data-centric analysis frameworks add the same message from a different angle (Puthankattil and Joseph, 2012; Ghiasi et al., 2021; Zhao et al., 2022). Leakage-specific and data-centric critiques make the underlying design problem explicit (Shim et al., 2021; Shen et al., 2025).

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That tension is especially sharp in relatively small neuromodulation studies. Rich models can improve representation learning, but relatively small cohorts also increase the risk of unstable feature selection, inflated performance estimates, and overly confident biomarker claims. Nested evaluation and within-fold processing are therefore central design issues rather than implementation details in this literature (Widge et al., 2019; Shim et al., 2021). Clinically deployable EEG biomarkers also need interpretability and low-density feasibility, and recent data-centric frameworks have started to move in that direction (Shen et al., 2025). Emerging response-prediction studies in depression and related neuromodulation settings show substantial promise (Funk and George, 2008; Bailey et al., 2023; Murphy et al., 2023). Related reviews and pilot studies also make the methodological fragility clear (Guo et al., 2024; Tsai et al., 2023; Dagnino et al., 2023). More recent response-oriented studies continue that pattern (Sheen et al., 2024; Zeidabadi and Rashidi, 2025; Reissmann et al., 2026). A recent systematic review of EEG connectivity changes in depression treatment reaches a similar conclusion (Ulrich et al., 2025). Within the same home-based 4-channel MDD program studied here, recent PSD-based and PLV-based analyses have already shown that the baseline EEG carries treatment-response information across multiple signal representations (Moncy et al., 2024; Xiao et al., 2025b). That background makes this cohort a useful setting for testing whether a sparse interpretable feature-selection workflow can isolate a small recurrent biomarker set under strict participant-level validation.

This manuscript focuses on a narrow objective: sparse biomarker discovery under strict participant-level evaluation. We compare two classifiers over the same feature-selection sweep to test whether a low-montage resting-state EEG pipeline can recover a small recurring set of candidate markers associated with tDCS response in adults with TRD. The emphasis is on parsimony, candidate-biomarker interpretability, and leakage-resistant validation. Our working hypothesis was that useful patient-level discrimination could be achieved with a small per-fold feature budget and that the most recurrent features would show stronger effect-direction consistency than hard-set overlap.

2. Methods

2.1. Dataset and preprocessing

The analysis used pretreatment EEG from 21 adults with treatment-resistant depression enrolled in a fully remote, multisite, double-blind, randomized sham-controlled home-based tDCS trial for major depressive disorder (Woodham et al., 2025). The analyzed cohort contained 7 remission cases and 14 non-remission cases in the patient-level outcome labels, with 18 female participants and a mean age of 37.1 ± 9.7 years. Table 1 summarizes the cohort, acquisition, and sweep configuration.

Baseline recordings were collected during eyes-closed rest with a 4-channel montage (AF7, AF8, TP9, TP10) sampled at 256 Hz. The available study exports comprised 10 s segments with 2,560 samples per channel, and the underlying recording sessions were stored either as a single 10 min file or as two 5 min files depending on participant. Across the cohort, the export set contained 1,203 windows, including 393 remission-labeled and 810 non-remission-labeled windows. This acquisition setup uses a limited frontal-temporal montage with clear portable-deployment relevance (Casson et al., 2010; Stoyell et al., 2021). The montage also constrained the downstream feature space toward frontal, temporal, cross-hemispheric, and frontal-temporal interaction measures.

For the present sweep, the processing configuration applied participant-level per-channel demeaning before upsampling, followed by a fourth-order Butterworth low-pass filter at 60 Hz, a downsampling step back to 256 Hz, and zero-overlap window slicing. The sweep then compared non-overlapping analysis windows of 2, 4, 6, 8, and 10 s, so the model inputs included both the native 10 s exports and shorter subdivisions derived from the same recordings. These choices materially affect downstream EEG feature distributions, especially in low-density or portable pipelines (Siuly et al., 2017; Miranda et al., 2019; Mahato et al., 2020). They also intersect with standardization and signal-quality concerns in wearable EEG settings (BABILONI et al., 2012; Simmatis et al., 2023). The methodological contribution of the present sweep lies in combining this broad handcrafted pool with participant-level outer validation, fold-internal feature selection, fold-internal class rebalancing, and post hoc recurrence summaries, rather than in introducing a single novel preprocessing transform (Widge et al., 2019; Shim et al., 2021; Shen et al., 2025).

2.2. Feature extraction

Feature extraction was performed on each preprocessed window. With all four channels available, the extractor generated 247 EEG-derived measures per window spanning channel-local descriptors, regional asymmetry terms, frontal-temporal interaction terms, synchrony summaries, and a small set of TP10-specific derived measures. Each channel contributed 39 local measures, comprising 21 spectral descriptors, 12 time-domain and Hjorth descriptors, and 6 entropy or nonlinear complexity descriptors, for 156 channel-local measures overall. The remaining 91 measures summarized asymmetry, frontal-temporal balance, cross-channel synchrony, and TP10-focused derived descriptors. This broad feature basis was deliberate. The goal of the study was to compare feature families under a leakage-safe sparse-selection regime rather than to pre-commit the analysis to a single biomarker hypothesis (Zhang et al., 2023; Shen et al., 2025). Table 2 summarizes the extracted feature families and their computational rationale.

Table 1

Cohort, acquisition, and sweep configuration summary. Trial descriptors derive from the parent home-based tDCS study (Woodham et al., 2025); acquisition totals and sweep settings derive from the analyzed EEG exports and current experiment configuration.

Characteristic	Value
Parent trial and cohort descriptors	
Parent trial	Fully remote, multisite, sham-controlled home-based tDCS trial for MDD (Woodham et al., 2025)
Participants	21
Outcome groups	7 remission, 14 non-remission
Sex (female/male)	18/3
Age (years, mean $\pm$ SD)	37.1 $\pm$ 9.7
EEG acquisition and export descriptors	
Rest condition	Eyes-closed resting state
EEG montage	4 channels (AF7, AF8, TP9, TP10)
Channel coverage	Frontal (AF7, AF8), temporal (TP9, TP10)
Sampling rate	256 Hz
Recording structure	10 min total; single file or two 5 min files
Available export segment length	10 s (2,560 samples/channel)
Total exported windows	1,203
Remission windows	393 (32.7%)
Non-remission windows	810 (67.3%)
Class ratio	2.06:1 (non-remission:remission)
Current sweep methodology settings	
Sweep window sizes	2, 4, 6, 8, 10 s
Inner-k sweep	1, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70
Feature selector	SelectKBest (f -classif)
Training adjustments	LOPO-group equalization + SMOTE
Validation	Leave-one-participant-out
Evaluation level	Patient-level aggregation

Resting-State EEG Processing and Evaluation Pipeline

All selection and class rebalancing steps are confined to the training folds before held-out participant scoring.

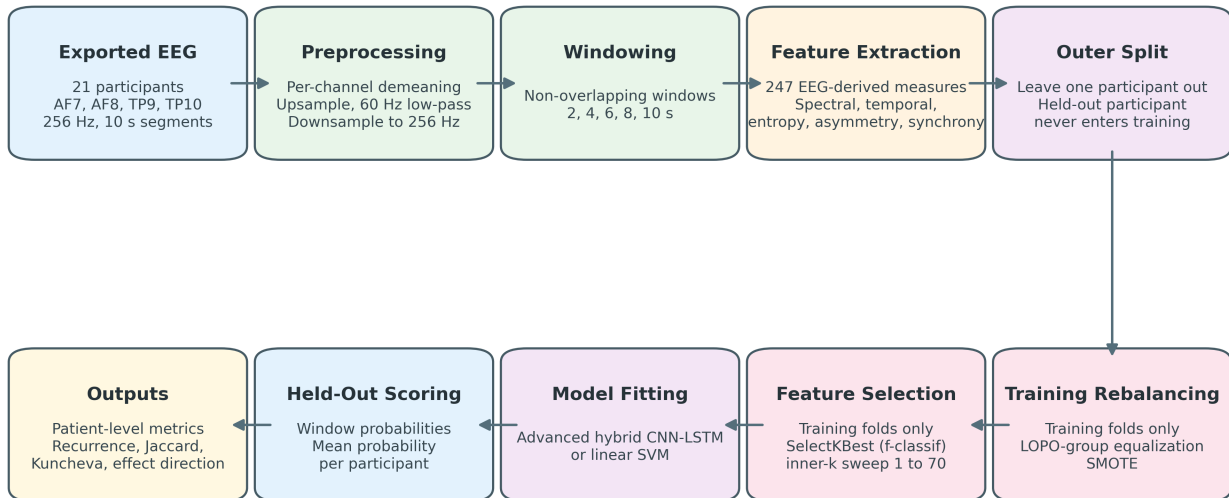


Figure 1: End-to-end analysis flow used in the manuscript. Participant separation, rebalancing, and feature selection were all applied inside the training folds before held-out participant scoring and biomarker summarization.

Table 2

Summary of extracted EEG feature families per analysis window.

Feature family	Measures	Count	Why included
Channel-local spectral	Absolute and relative delta, theta, alpha, beta, and gamma power; total power; frequency-weighted power; SEF90 and SEF95; PSD mean, standard deviation, maximum, minimum, median, skewness, and kurtosis	21/channel (84 total)	Canonical resting-state EEG descriptors with clear band-level interpretation and strong precedent in depression EEG
Channel-local time domain and Hjorth	Mean, standard deviation, variance, skewness, kurtosis, root-mean-square amplitude, peak-to-peak range, zero crossings, mean absolute deviation, Hjorth activity, mobility, and complexity	12/channel (48 total)	Captures amplitude dispersion, oscillatory rate, roughness, and short-window morphology with low computational cost
Channel-local entropy and nonlinear complexity	Sample entropy, spectral entropy, correlation dimension, Hurst exponent, Lyapunov exponent, and detrended fluctuation analysis	6/channel (24 total)	Captures irregularity and complexity beyond band-power summaries
Band-power asymmetry and frontal-temporal interactions	Frontal and temporal asymmetry, hemispheric asymmetry, ipsilateral frontal-temporal ratios and normalized differences, and diagonal AF7-TP10 and AF8-TP9 interactions across frequency bands	70 total	Tests laterality and frontal-temporal balance within the four-channel montage
Synchrony and coupling	Pearson correlation and maximum normalized cross-correlation over six anatomical channel pairs, plus legacy cross-correlation summaries	18 total	Summarizes short-window coupling and cross-channel coordination
TP10-specific derived descriptors	Peak prominence, peak concentration ratio, and an asymmetry-weighted peak-prominence term	3 total	Targets concentrated temporal spectral structure with straightforward interpretation
Total EEG-derived features	Combined feature space per window	247 total	Broad hypothesis space for leakage-safe sparse selection

For each of AF7, AF8, TP9, and TP10, we computed 39 channel-local features. Spectral features were estimated from Welch power spectral densities (Welch, 1967) and included absolute band power in the delta, theta, alpha, beta, and gamma ranges, total power, relative band power normalized by total power, frequency-weighted power, spectral edge frequencies at 90% and 95%, and PSD summary statistics (mean, standard deviation, maximum, minimum, median, skewness, kurtosis). These measures were included because spectral content remains one of the most widely used and interpretable feature families in depression EEG work (Olbrich and Arns, 2013; Mumtaz et al., 2017; Mahato et al., 2020). Temporal-channel spectral content has also appeared repeatedly in pretreatment response studies (Jaworska et al., 2018).

Time-domain descriptors were also computed for each channel. These included mean, standard deviation, variance, skewness, kurtosis, root-mean-square amplitude, peak-to-peak range, zero crossings, and mean absolute deviation, together with Hjorth activity, mobility, and complexity (Hjorth, 1970; Zhang et al., 2023). These features summarize amplitude dispersion, local oscillatory rate, signal roughness, and short-window morphology. They are computationally light and well suited to low-montage resting-state EEG, where clinically useful features must remain interpretable and robust under short windows.

Entropy and nonlinear complexity measures were computed per channel to capture irregularity beyond simple power summaries. The extractor included sample entropy, spectral entropy, correlation dimension, Hurst exponent,

Lyapunov exponent, and detrended fluctuation analysis. Complexity measures were calculated on a demeaned, standardized version of each signal, with long windows subsampled for numerical stability. These descriptors were retained because depression EEG studies have repeatedly implicated entropy, variability, and broader complexity structure (Puthankattil and Joseph, 2012; Faust et al., 2014; Acharya et al., 2015). That pattern also appears in pretreatment response and large-scale resting-state evaluations, even when the exact direction of effect depends on cohort, montage, and temporal scale (Jaworska et al., 2018; Zhao et al., 2022; Lord and Allen, 2023).

The four-channel montage also supports a range of regional interaction features derived from cached band-power estimates. Traditional frontal asymmetry (AF7/AF8) and temporal asymmetry (TP9/TP10) were computed for each frequency band in two forms: a log-ratio index,

$$AI = \log\left(\frac{P_R}{P_L}\right),$$

and a normalized difference,

$$ND = \frac{P_R - P_L}{P_R + P_L}.$$

Using the same band-power cache, we also computed hemispheric asymmetry by averaging AF7+TP9 and AF8+TP10 within each hemisphere, ipsilateral frontal-temporal ratios and normalized differences on the left and right hemispheres, and diagonal frontal-temporal interactions

for AF7-TP10 and AF8-TP9. These measures were included because reduced-channel depression EEG studies repeatedly point to laterality, temporal involvement, and frontal-temporal balance rather than to a single isolated electrode (Mahato et al., 2020; Ghiasi et al., 2021; Monni et al., 2022).

Pairwise synchrony features were added for six anatomically meaningful channel pairs: AF7-AF8, TP9-TP10, AF7-TP9, AF8-TP10, AF7-TP10, and AF8-TP9. For each pair we computed maximum normalized cross-correlation and Pearson correlation, and the extractor also retained a legacy set of raw cross-correlation summaries across all channel pairs. These features were intended to capture short-window synchrony and coupling patterns that have been emphasized in resting-state depression EEG and neuromodulation response work (Ghiasi et al., 2021; Zhao et al., 2022; Ulrich et al., 2025).

Finally, the extractor included three exploratory TP10-specific derived descriptors: peak prominence, peak concentration ratio, and an asymmetry-weighted peak-prominence term. These summarize whether the TP10 spectrum contains a sharp dominant peak, how concentrated that peak is relative to total power, and whether that concentration covaries with ipsilateral frontal-temporal beta balance. They were included because temporal-channel spectral content has recurrently appeared in low-montage depression EEG analyses, while still remaining interpretable enough to survive foldwise sparse selection if informative (Mumtaz et al., 2017; Mahato et al., 2020).

2.3. Sweep design and validation

The experiment sweep compared two model families over the same feature-selection protocol: an advanced hybrid 1D CNN-LSTM and a linear SVM. All runs used sequential window ordering, SelectKBest with ANOVA F statistics, leave-one-participant-out outer evaluation, leave-one-participant-out training-group equalization, and SMOTE within training folds (Chawla et al., 2002). The sweep varied the per-fold feature-selection budget (inner-k) across 15 values from 1 to 70 and repeated the search across 5 window sizes, yielding 150 successful runs in total (75 per model).

This design matters for interpretation. The outer evaluation remained subject independent throughout, so no participant contributed both training and test windows to the same outer fold. Feature selection, participant-group equalization, and synthetic oversampling were all restricted to the outer training folds, which is important because leakage at these steps can materially inflate reported biomarker performance (Shim et al., 2021). The inner-k parameter controlled how many features were selected inside each outer training fold. The feature-selection stability panels in the manuscript describe shared pipeline behavior because both classifiers consumed the same selected features for matched sweep settings.

2.4. Models

Both classifiers operated on the fold-selected handcrafted feature vectors produced by the current inner-k setting. Patient-level outputs were derived by averaging held-out window probabilities within each participant before computing ROC-AUC and the other clinical metrics. Table 3 summarizes the model architecture and training setup used in the sweep.

The SVM baseline used a linear kernel with $c=1.0$ and probability output enabled. When SMOTE was active inside the training folds, class weighting was not applied in parallel, so synthetic oversampling and explicit class weighting were not used simultaneously.

The hybrid model treated the selected feature vector as a learned one-dimensional signal before recurrent modeling. In the best-performing 6 s, $\text{inner-k}=1$ run, the input features were first projected to a 128-dimensional embedding with normalization and dropout, then mapped into a 64-step latent sequence for three convolutional blocks. These blocks used channel progressions of [64, 64], [128, 128], and [256, 128] with kernel sizes [5, 3], [3, 3], and [3, 1], together with batch normalization, max pooling, and dropout after each stage. The resulting convolutional sequence was passed directly to the recurrent stage rather than being averaged beforehand, so sequence structure remained available to the LSTM layers.

Sequence modeling then used two bidirectional LSTM layers with 128 and 64 hidden units, followed by eight-head self-attention with residual layer normalization and dropout. The recurrent summary was globally pooled and concatenated with pooled multi-scale CNN outputs through feature-pyramid fusion before two dense blocks of 256 and 128 units and a final two-logit output layer. Training used AdamW, cosine warm restarts, label smoothing, gradient clipping, mixed precision, and early stopping with best-state restore.

The architectural choices were intended to distribute the modeling burden across complementary stages. The projection layer compressed the sparse selected features into a stable latent representation before deeper processing. The convolutional front end summarized short-range interactions and produced progressively richer multi-scale feature maps. The bidirectional LSTM layers then modeled ordered dependencies across that latent sequence, while self-attention allowed the model to reweight informative positions after recurrent encoding. Feature-pyramid fusion retained both shallower and deeper convolutional summaries, which is useful when the most discriminative structure may not reside at a single representational depth. The dense head provided a final nonlinear compression stage before binary classification. Batch normalization, dropout, label smoothing, gradient clipping, and early stopping were included to stabilize optimization and limit overfitting in a relatively small cohort.

2.5. Outcome metrics and biomarker summaries

The primary endpoint was patient-level ROC-AUC, computed from participant probabilities aggregated

Table 3

Model architecture and training summary for the two evaluated classifier families.

Component	Configuration
Input to both models	Fold-selected handcrafted feature vector from the current <i>inner-k</i> setting. Patient-level predictions were computed by averaging held-out window probabilities within each participant.
SVM baseline	Linear-kernel SVM with $C=1.0$ and probability estimates enabled. When SMOTE was active in the training folds, class weighting was disabled to avoid double balancing.
Hybrid input embedding	Linear projection from the selected feature vector to 128 latent units, followed by normalization, a nonlinear activation, dropout 0.1, and remapping to a learned one-dimensional sequence of length 64.
CNN trunk	Three 1D convolutional blocks with filter progressions [64,64], [128,128], and [256,128]; kernel sizes [5,3], [3,3], and [3,1]; max-pooling factors 2, 2, and 1; batch normalization; and dropout [0.1, 0.2, 0.3].
Sequence stage	The convolutional output sequence was passed directly to the recurrent stack. Two bidirectional LSTM layers used 128 and 64 hidden units.
Attention and fusion	Eight-head self-attention with dropout 0.1 and residual layer normalization, followed by global mean pooling. Multi-scale pooled CNN features were concatenated through feature-pyramid fusion.
Dense head	Two dense blocks of 256 and 128 units with batch normalization, nonlinear activations, and dropout 0.3 and 0.4, followed by a two-logit output layer.
Optimization	AdamW with learning rate 0.001 and weight decay 0.01; cosine warm restarts to a minimum learning rate of 10^{-6} ; batch size 8192 before fold-size adjustment; label smoothing 0.05; gradient clipping at 1.0; mixed precision; maximum 100 epochs; early stopping patience 5 with best-state restore.

over held-out windows. Secondary patient-level metrics included PR-AUC, balanced accuracy, accuracy, F1, MCC, precision, recall, and specificity. The pipeline also reported bootstrap confidence intervals and a permutation test for the patient-level ROC-AUC summary. All main manuscript claims use patient-level metrics. Window-level scores appear only as intermediate outputs.

To characterize biomarker behavior across outer folds, we used several post hoc summaries: mean pairwise Jaccard overlap, Kuncheva index, unique feature count, top-feature share, class-conditional selection deltas, and effect-direction consistency. These summaries are descriptive. They characterize recurrence and directionality across folds and do not support causal interpretation. Low overlap can coexist with recurring features that retain consistent class-separation direction when selected (Olbrich and Arns, 2013; Olbrich et al., 2015; Simmatis et al., 2023).

3. Results

3.1. Sweep-level comparison

Across the 150 successful runs, the hybrid model outperformed the SVM baseline on the main patient-level discrimination metrics over most of the sweep landscape. The performance advantage was visible when averaging over window sizes (Figure 2) and when averaging over *inner-k* settings (Figure 3). The best individual hybrid run occurred at 6 s windows with *inner-k*=1, whereas the best SVM run occurred at 8 s windows with *inner-k*=30.

Figure 3 shows that the relationship between ROC-AUC and *inner-k* differed across the two model families in the low-budget region. The hybrid model retained clearly stronger ROC-AUC across the sparsest budgets and achieved the highest individual ROC-AUC in the entire sweep at *inner-k*=1. The linear SVM did not show the same pattern and reached its best ROC-AUC only at

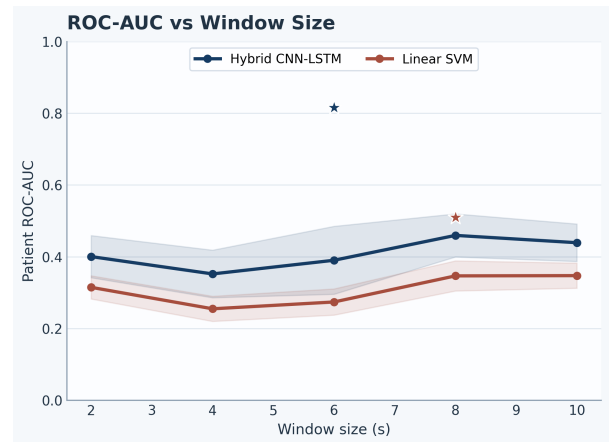


Figure 2: Patient-level ROC-AUC by window size across the sweep. Stars mark the best individual hybrid and SVM runs.

the broader *inner-k*=30 setting. Sparse per-fold selection therefore aligned with peak discrimination only for the more expressive model.

Figures 4 and 5 show a second related pattern. Increasing *inner-k* increased both hard-set overlap and the total number of unique selected features. Larger feature budgets did not produce the strongest patient-level discrimination. The sweep concentrated its best hybrid performance at the sparsest end of the per-fold selection range. This pattern motivates the paper's focus on sparse biomarker discovery and suggests that model capacity and per-fold sparsity mattered together rather than independently.

3.2. Best-run patient-level performance

The best hybrid configuration achieved patient-level ROC-AUC 0.816, PR-AUC 0.579, balanced accuracy

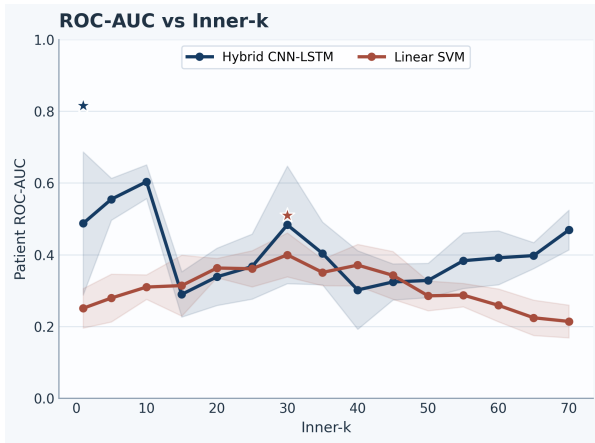


Figure 3: Patient-level ROC-AUC by inner-k across the sweep. The hybrid model retained the strongest discrimination in the sparsest budget region.

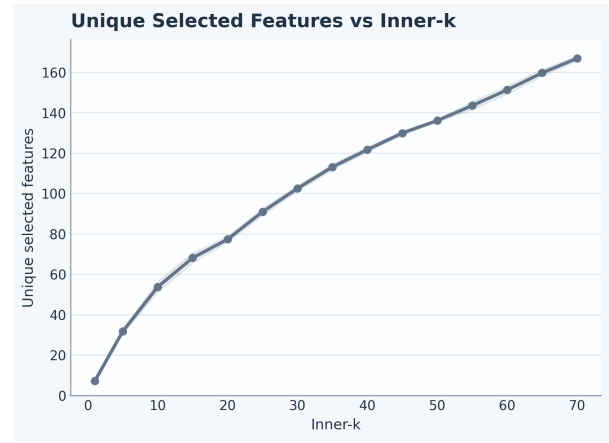


Figure 5: Unique selected features by inner-k. Larger per-fold budgets produced markedly larger discovered feature pools.

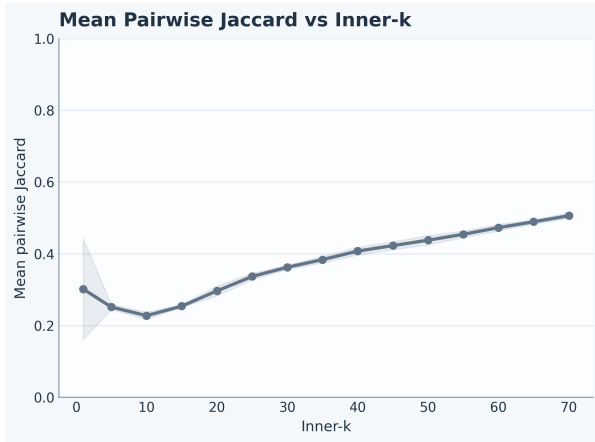


Figure 4: Mean pairwise Jaccard overlap by inner-k. Hard-set overlap increased with larger per-fold feature budgets.

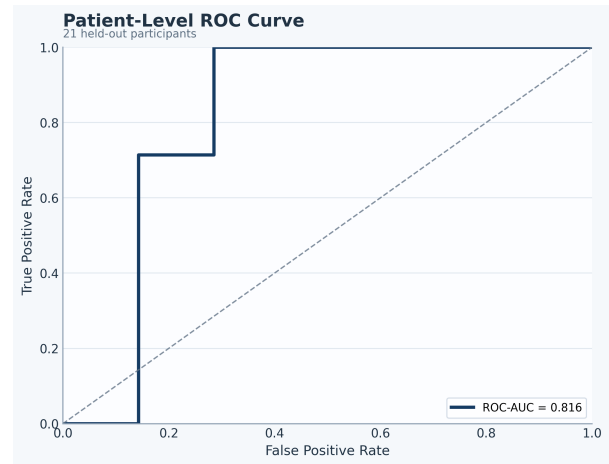


Figure 6: Patient-level ROC curve for the best hybrid run.

0.786, accuracy 0.810, F1 0.714, and MCC 0.571 over 21 held-out participants (Figures 6 and 7; Table 4). Its ROC-AUC bootstrap interval ranged from 0.600 to 0.990, and the permutation test remained significant ($p = 0.008$), which is notable given the relatively small cohort.

The corresponding confusion matrix contained 12 true negatives, 5 true positives, 2 false positives, and 2 false negatives (Figure 8). The metric summary in Figure 9 shows the same run in condensed form.

The best SVM baseline reached balanced accuracy 0.643 and ROC-AUC 0.510, with a non-significant permutation result ($p = 0.469$). This weaker performance came with a much larger per-fold feature budget (inner-k=30) and a far larger overall candidate set. The contrast indicates that the sparse hybrid pipeline extracted stronger patient-level discrimination from a much smaller subset.

3.3. Sparse biomarker set with modest hard-set overlap

The winning hybrid run selected an average of 1 feature per fold and yielded only 7 unique recurrent features across

the entire outer-validation process (Table 5). That degree of sparsity is one of the most distinctive findings in the sweep. Hard-set overlap was modest: mean pairwise Jaccard 0.190 and mean Kuncheva index 0.187. Figures 10 and 11 summarize the small recurrent pool together with the modest chance-adjusted stability summary.

A small recurring feature set emerged repeatedly, though not in every fold, and the selected set was dominated by temporal and frontal-temporal descriptors. The most recurrent feature was `tp10_spectral_entropy`, selected in 8 of 21 folds. The next most recurrent features were `left_frontal_temporal_diff_gamma`, `left_frontal_temporal_diff_beta`, and `af7_zero_crossings`.

3.4. Effect-direction consistency and class-conditional patterns

The recurrent hybrid biomarkers showed perfect mean sign consistency across the effect-stability summary. This supports treating them as promising biomarkers with stable directionality across the folds in which they were selected (Figure 12).

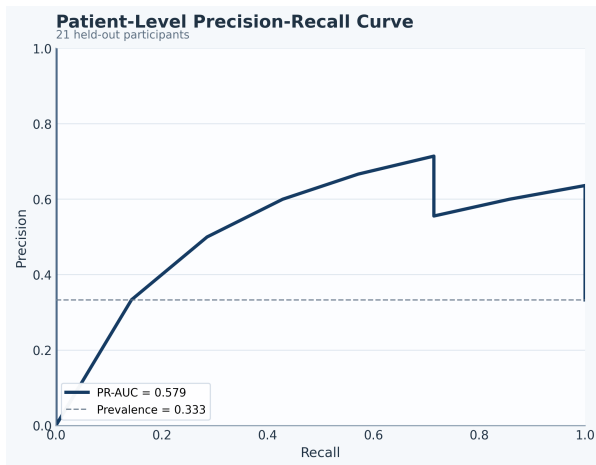
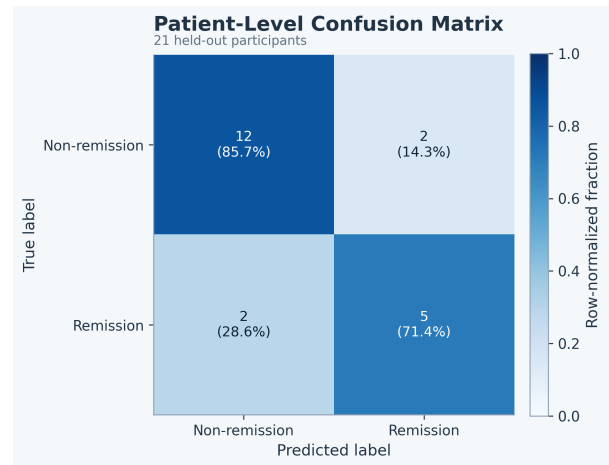
Table 4

Best-performing configuration for each model family.

Model	Window	Inner-k	ROC-AUC	PR-AUC	Bal. Acc.	F1	MCC	Feat./fold	Unique feat.	Mean Jaccard
Hybrid CNN-LSTM	6	1	0.816	0.579	0.786	0.714	0.571	1.0	7	0.190
Linear SVM	8	30	0.510	0.411	0.643	0.533	0.277	30.0	103	0.360

Table 5Recurrent biomarkers from the best hybrid run. Δ selection is remission minus non-remission selection share.

Feature	Folds	Share	Δ sel.	Sign cons.	Mean d	Effect direction
tp10_spectral_entropy	8	0.381	-0.571	1.000	-0.935	Non-remission > remission
left_frontal_temporal_diff_gamma	4	0.190	-0.071	1.000	-1.059	Non-remission > remission
left_frontal_temporal_diff_beta	3	0.143	0.000	1.000	-0.915	Non-remission > remission
af7_zero_crossings	3	0.143	0.429	1.000	0.814	Remission > non-remission
tp10_hurst	1	0.048	0.143	1.000	-1.010	Non-remission > remission
af7_tp10_diff_gamma	1	0.048	0.143	1.000	-1.203	Non-remission > remission
frontal_asymmetry_ratio_delta	1	0.048	-0.071	1.000	0.794	Remission > non-remission

**Figure 7:** Patient-level precision-recall curve for the best hybrid run.**Figure 8:** Patient-level confusion matrix for the best hybrid run.

tp10_spectral_entropy and the frontal-temporal gamma and beta difference terms were repeatedly associated with the non-remission group, whereas af7_zero_crossings was more frequently selected in remission folds. The class-conditional selection frequencies make that separation visible at the level of foldwise recurrence (Figure 13).

The effect-size summaries followed the same broad pattern, with af7_zero_crossings retaining a positive mean effect size and the main TP10 and frontal-temporal features retaining negative mean effects (Figure 14). This distinction matters clinically. The selection-frequency deltas do not justify definitive mechanistic claims. They provide a concrete interpretation target for follow-up validation. The best hybrid pipeline contributes two linked observations: useful patient-level discrimination can be achieved with an extremely sparse feature budget, and the resulting recurrent biomarkers show interpretable directionality even when fold-to-fold set overlap remains modest.

4. Discussion

The strongest patient-level discrimination in this sweep emerged from the sparsest tested configuration. The best hybrid run used inner-k=1, selected only 7 unique recurrent features overall, and exceeded the best linear baseline by a large margin on ROC-AUC and MCC. Figure 3 also shows that this was not a generic effect of smaller feature budgets across both models. In the low-budget region, the hybrid model retained stronger ROC-AUC than the linear SVM and reached its peak performance at the sparsest tested setting, whereas the linear model reached its best ROC-AUC only after a broader per-fold budget. In this cohort, the value of the more expressive model and the value of strong sparsity appear to have operated together.

One plausible explanation is that aggressive per-fold pruning removed weak or redundant handcrafted features and exposed a compact signal set whose interactions were more useful to the hybrid architecture than to a linear

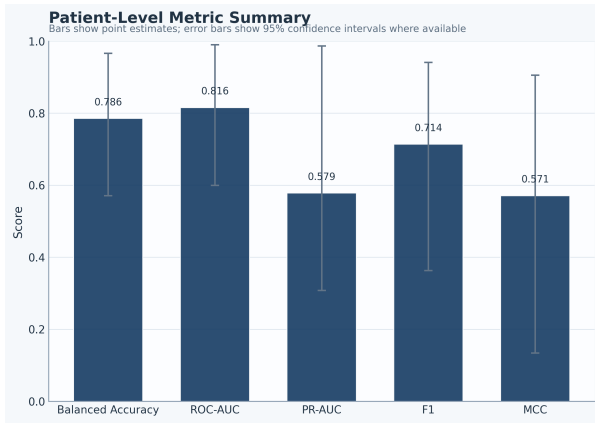


Figure 9: Patient-level metric summary for the best hybrid run.

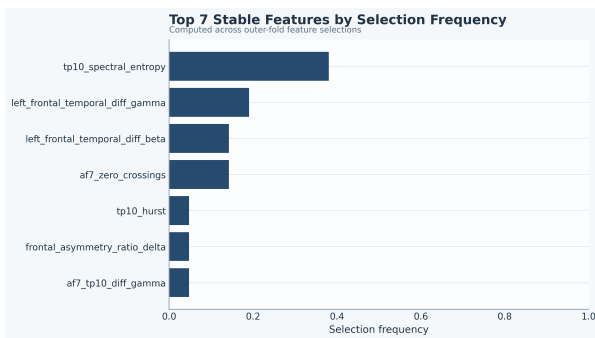


Figure 10: Selection-frequency ranking for the recurrent biomarkers from the best hybrid run.

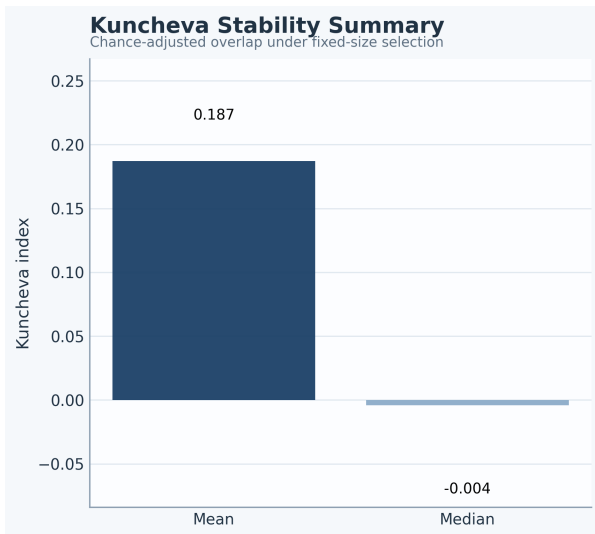


Figure 11: Kuncheva stability summary for the recurrent biomarkers from the best hybrid run.

decision boundary. With only a few retained descriptors, the convolutional, recurrent, and attention stages may have been better positioned to combine complementary temporal and frontal-temporal information without dilution from a larger pool of unstable candidates. The linear baseline appears to have depended more on accumulating a broader

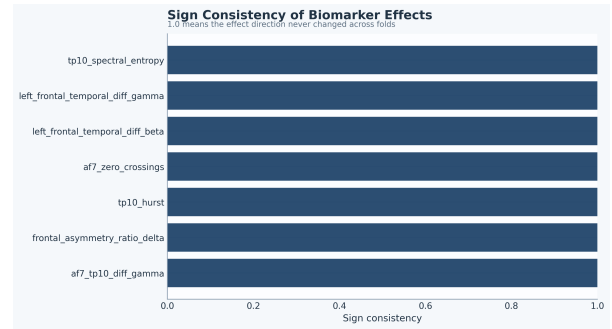


Figure 12: Sign-consistency summary for the recurrent biomarkers from the best hybrid run. All recurrent features retained the same effect direction across the folds in which they were selected.

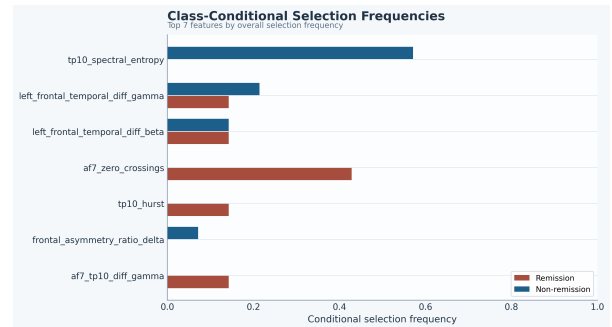


Figure 13: Class-conditional selection frequencies for the recurrent biomarkers from the best hybrid run. Remission is shown in red and non-remission in blue.

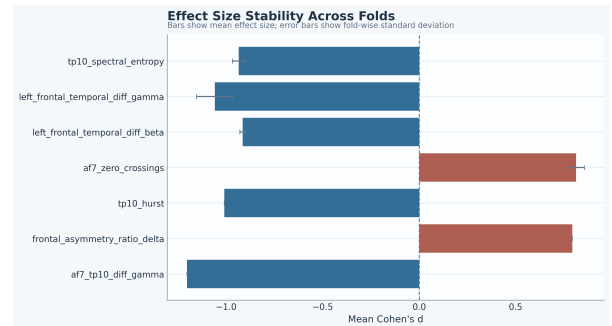


Figure 14: Effect-size stability for the recurrent biomarkers from the best hybrid run. Positive values indicate remission-skewed effects and negative values indicate non-remission-skewed effects.

set of moderate additive cues before reaching its best ROC-AUC. This interpretation remains provisional, but it is consistent with the sweep pattern and with the marked separation between the best sparse hybrid run and the broader-budget linear baseline.

This result fits part of the EEG depression literature and supports a measured interpretation. Prior work has argued that EEG can provide clinically useful baseline markers in major depressive disorder and treatment-response settings (Olbrich and Arns, 2013; Olbrich et al., 2015; Leuchter et al., 2010). Related discussions have also emphasized

tighter biomarker discipline, better standardization, and more realistic expectations about generalization (Simmatitis et al., 2023; Alonso et al., 2017; Fu et al., 2019). Machine learning studies have likewise reported encouraging treatment-response signals, including combinations of EEG and clinical covariates (Al-Kaysi et al., 2017; Jaworska et al., 2019; Oakley et al., 2022). Other modelling studies reach similar conclusions while still exposing limits from data heterogeneity, cohort size, or insufficient interpretability (Shalbaf et al., 2018; Ebrahimzadeh et al., 2021; Zhdanov et al., 2020). Recent predictive overviews and reviews continue to highlight the same translational tension (Watts et al., 2022; Cohen et al., 2023).

The present sweep adds a feature-selection view to that literature. Figures 4 and 5 show that larger feature budgets yield more overlap and many more unique discovered features. Those larger budgets do not improve discrimination. The panels support a small recurring biomarker set with consistent direction of class separation. That reading keeps the modest Jaccard and Kuncheva scores in view and gives appropriate weight to the stronger effect-direction signal. Methodologically, this framing matters because the depression EEG biomarker literature still contains substantial publication bias, limited out-of-sample validation, and known vulnerability to leakage from pre-cross-validation feature processing (Widge et al., 2019; Shim et al., 2021). The present pipeline addresses that problem by embedding selection and rebalancing inside participant-level outer folds and then summarizing recurrence and directionality instead of reporting only a single global feature list (Shen et al., 2025).

The same clinical program has also supported other signal representations in recent PSD-based and PLV-based studies, which located informative response-related signal in low-density temporal and frontal-temporal patterns (Moncy et al., 2024; Xiao et al., 2025b). In the present sweep, the recurrent feature set again concentrated in TP10-derived and frontal-temporal descriptors. That regional convergence supports the view that the low-montage recordings contain response-relevant information across multiple feature families.

The specific biomarkers in the present sweep are plausible enough to deserve follow-up. `tp10_spectral_entropy` was the most frequently selected feature and skewed toward non-remission-associated folds. Entropy and related nonlinear descriptors have appeared repeatedly in depression EEG studies (Puthankattil and Joseph, 2012; Faust et al., 2014; Acharya et al., 2015). Pretreatment multiscale entropy and signal-variability measures have also been linked to antidepressant treatment response, although the reported direction depends on temporal scale and recording context (Jaworska et al., 2018). That nuance is important because a recent large resting-state evaluation found complexity metrics to be stable but not to support a simple uniform depression-complexity association (Lord and Allen, 2023). The present TP10 entropy finding therefore fits a broader complexity literature

and supports a credible temporal biomarker signal in this cohort, while still needing external confirmation.

The frontal-temporal beta and gamma difference features also have prior support at the family and regional level. Temporal electrodes, higher-frequency band features, and paired asymmetry terms have been reported as informative in depression EEG studies, and pretreatment frontal-temporal features have been linked to treatment outcome in MDD (Mahato et al., 2020; Mumtaz et al., 2017). The current beta and gamma difference terms are therefore better read as family-level extensions of that regional pattern than as exact replications of a previously fixed biomarker.

`af7_zero_crossings` had the clearest remission-skewed selection pattern in the current sweep, but it has less direct precedent as a named biomarker. AF7-centered signal relevance has appeared in adjacent low-montage work (Moncy et al., 2025), and broader frontal EEG literature continues to implicate asymmetry and signal-complexity families in depression (Monni et al., 2022; Lord and Allen, 2023). The specific zero-crossing formulation used here is therefore best regarded as a plausible frontal biomarker in this setting and should be tested in independent cohorts. Taken together, the recurrent set emphasized temporal and frontal-temporal descriptors, with limited concentration in any single frontal asymmetry family. This regional profile is compatible with the broader low-montage findings discussed above. The current study contributes a sparse handcrafted biomarker signature with explicit recurrence and effect-direction summaries.

Related home-based tDCS studies in bipolar depression from the same group point in a similar low-montage direction. PSD-based modeling again highlighted AF7 and TP10 signal content, and PLV analyses emphasized frontal-temporal and temporoparietal synchronization patterns (Moncy et al., 2025; Xiao et al., 2025a). These bipolar studies address a different diagnosis and treatment protocol. They support reduced-montage feasibility and signal plausibility in adjacent affective cohorts.

The dataset is relatively small ($n = 21$), the confidence intervals are wide, and there is no external validation set. The bottom-row sweep panels describe the selection pipeline itself because the same feature-selection layer feeds both classifiers. Class-conditional ratios also become unstable when a feature is absent from one class, so the manuscript uses deltas and effect direction as the main descriptive summaries. The recurring features can be treated as plausible biomarkers from this cohort that now warrant follow-up validation.

Future work should test whether the same sparse feature set recurs under independent cohorts, alternative low-density montages, and explicit external validation (Dagnino et al., 2023). Adjacent response-prediction work in depression and neuromodulation provides useful targets for that next stage (Funk and George, 2008; Bailey et al., 2023; Guo et al., 2024). Home-based and transport-oriented studies offer one clear direction for

follow-up (Romero Marin et al., 2026; Kratter et al., 2026; Waller et al., 2026). More recent response-signature studies also provide relevant benchmarks for that next phase (Sheen et al., 2024; Zeidabadi and Rashidi, 2025; Reissmann et al., 2026). Related protocol and review work in treatment-resistant depression can help frame that validation effort (Ulrich et al., 2025; Pettorruso et al., 2024). These extensions should keep stringent patient-level evaluation and sparse interpretable feature budgets in the core study design.

5. Conclusion

In this sweep, low-montage resting-state EEG supported useful patient-level discrimination of tDCS response in adults with treatment-resistant depression, and the strongest result came from an extremely sparse feature-selection regime. The best hybrid run achieved ROC-AUC 0.816 while selecting only 1 feature per fold and 7 unique recurrent features overall. Hard-set overlap across folds remained modest. The present evidence supports a small, interpretable biomarker signature with consistent effect direction under strict participant-level validation.

That combination of parsimony, interpretability, and subject-independent evaluation defines the main contribution of the present draft. External validation in larger cohorts is the next requirement. The current results support the feasibility of sparse short-timescale EEG biomarker discovery for tDCS response in a clinically realistic low-density setting.

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